

MITHULRAM GUNASEKARAN

Software Engineer | Backend, Data & Secure Systems

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Software engineer and M.Sc. Computer Science student with 3 years of software-development experience across Java, Python, REST APIs, mobile-to-backend workflows, and data handling. Improved user engagement by 20% and data-handling efficiency by 3x through product delivery at Xmplar. Adds applied automotive-cybersecurity research experience in threat modeling, risk assessment, vulnerability documentation, and technical reporting. Builds tested, understandable systems where access decisions, data quality, and operational behavior are explicit.

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, C++, SQL, Dart

Backend & Data: Spring Boot, Spring Security, FastAPI, DuckDB, JPA, REST APIs, data quality, API integration

Frontend & Mobile: React, Flutter, Firebase Analytics, Crashlytics, notifications, deep links

Cloud & Delivery: Microsoft Azure, Git, GitLab, GitHub Actions, CI/CD, technical documentation

Security & Systems: threat modeling, risk scoring, vulnerability analysis, authentication, access control, ECU communication

PROFESSIONAL EXPERIENCE

Student Research Assistant | Automotive Cybersecurity

University of Passau (ASOA) | Passau, Germany | Mar 2025 - Mar 2026

- Reviewed 20+ automotive-cybersecurity papers, standards, and technical reports to identify authentication gaps, insecure service discovery, denial-of-service vectors, and ECU communication failure modes.
- Documented 15+ vulnerability and failure-mode patterns; prioritized 10+ security and safety risks using likelihood-impact scoring, threat modeling, and distributed ECU communication analysis.
- Evaluated message integrity, access control, segmentation, secure updates, and attack-surface reduction; converted findings into structured technical documentation.

Junior Developer

Xmplar | Bangalore, India | Dec 2021 - Aug 2024

- Built responsive Flutter interfaces and backend-integrated workflows for a service platform, contributing to a 20% improvement in user engagement.
- Improved data-handling efficiency by 3x by refining processes and mobile-to-backend API data flows.
- Implemented Firebase Analytics, Crashlytics, notifications, deep links, and dynamic links; referral flows contributed to a 50% increase in engagement.

Backend Intern

Adpay | Chennai, India | May 2020 - Jul 2020

- Developed and documented Java and Spring Boot APIs for an Android distributor application.

SELECTED PROJECTS

Secure Asset Access API | Java, Spring Boot, Spring Security, JPA, REST APIs, GitHub Actions

github.com/mithulram/secure-asset-access-api

- Built a Java/Spring Boot API with validated REST endpoints, role-based authorization, JPA persistence, an audit grant registry, health checks, and end-to-end MockMvc security tests.
- Designed clear failure behavior with payload validation, role-boundary checks, audit records, and RFC 9457-style problem details for unknown assets.

Data Quality & Lineage Pipeline | Python, DuckDB, SQL, CSV, Parquet, GitHub Actions

github.com/mithulram/data-quality-lineage-pipeline

- Built a Python/DuckDB pipeline that validates order batches, quarantines invalid records, enriches valid data with customer dimensions, exports CSV and Parquet, and emits auditable lineage and quality-report artifacts under a configurable error-rate gate.
- Implemented reproducible rule reporting and quality-gate failure behavior so invalid input is visible rather than silently dropped.

EDUCATION

M.Sc. Computer Science

University of Passau | Passau, Germany | 2025 - Present

B.E. Computer Science and Information Technology

SRM University | Chennai, India | 2018 - 2022 | CGPA: 7.8/10

LANGUAGES

Tamil (Native) | English (Fluent) | German (B1 - Intermediate)

CERTIFICATIONS

Professional: Microsoft Azure Solutions Architect Expert (Microsoft, 2024) | Microsoft Azure Administrator AZ-104 (Microsoft, 2024) | Java SE 11 Developer (Oracle, 2022) | PCAP - Python Certified Associate (Python Institute, 2023) | Advanced Programming in C++ (C++ Institute, 2021)

Additional: Java Programming (HackerRank, 2023) | JavaScript Programming (HackerRank, 2023) | Python Programming (HackerRank, 2023) | SQL Programming (HackerRank, 2023) | Google Analytics (LinkedIn Learning, 2023)